

# André M. Silva

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## Short Bio

I am André Silva, a postdoctoral researcher at the Instituto de Astrofísica e Ciências do Espaço. My research spans instrumentation and exoplanetary science. I am currently working on the PoET solar telescope, leading the development of its observation software. In parallel, I focus on the development of novel algorithms for precise radial velocity extraction and analyze systematic effects that impact such measurements. Over the past three years, I have lectured an advanced course on Python, a Master's-level course on Bayesian statistics, and a PhD-level module on high-resolution spectroscopy.

## Education

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| <b>PhD</b> | <b>Faculdade de Ciências</b> , Doctor's Degree in Astronomy <ul style="list-style-type: none"><li>Thesis - A new paradigm for the estimation of precise stellar radial velocities</li></ul> | Universidade do Porto<br>2019 – 2024 |
| <b>MSc</b> | <b>Faculdade de Ciências</b> , Mestrado Integrado em Engenharia Física <ul style="list-style-type: none"><li>Thesis - An expansion to the CHEOPS mission official pipeline.</li></ul>       | Universidade do Porto<br>2014 – 2019 |

## Awards

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| <b>IAU PhD Prize 2024</b> <a href="#">🔗</a> , Division B                                                                                                                                                                        | 2025 |
| <ul style="list-style-type: none"><li>Awarded the PhD Prize 2024 from the International Astronomical Union (IAU) Division B (Facilities, Technologies, and Data Science) for the work developed during my PhD thesis.</li></ul> |      |
| <b>EAS MERAC 2026</b> <a href="#">🔗</a> , Division B                                                                                                                                                                            | 2025 |
| <ul style="list-style-type: none"><li>Awarded the European Astronomical Society (EAS) MERAC Prize in Observational Astrophysics for the work developed during my PhD thesis.</li></ul>                                          |      |

## Experience

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| <b>Instituto de Astrofísica e Ciências do Espaço</b> , Post-doctoral researcher <ul style="list-style-type: none"><li>Development of the observational software for the PoET solar telescope.</li></ul>                   | CAUP<br>Nov 2024 – present                 |
| <b>Faculdade Ciências da Universidade do Porto</b> , Invited Assistant Professor <ul style="list-style-type: none"><li>Invited assistant Professor to teach a PhD-level unit on high-resolution spectroscopy</li></ul>    | Physics department<br>Nov 2024 – present   |
| <b>Faculdade de Ciências da Universidade de Lisboa</b> , Post-doctoral researcher <ul style="list-style-type: none"><li>Development of software to measure day-time seeing, under the FIERCE ERC project</li></ul>        | FCiências.ID<br>Aug 2024 – Nov 2024        |
| <b>Faculdade Ciências da Universidade do Porto</b> , Invited assistant <ul style="list-style-type: none"><li>Invited assistant for the practical classes of MSc-level unit on Bayesian data analysis.</li></ul>           | Physics department<br>Feb 2024 – Sept 2024 |
| <b>Instituto de Astrofísica e Ciências do Espaço</b> , PhD fellow <ul style="list-style-type: none"><li>FCT funding for the PhD thesis "A new paradigm for the estimation of precise stellar radial velocities"</li></ul> | Physics department<br>2019 – July 2024     |

## Talks

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| <b>EAS2026</b> , On the extraction of precise radial velocities from high-resolution spectra - avenues and challenges | July 2026 |
| <b>exoplanets6</b> , Intrinsic limitations to template-based radial velocity extraction from high-resolution spectra  | June 2026 |

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| <b>EPRV x Solar Physics workshop</b> <a href="#">↗</a> , Looking at the Sun through ESPRESSO - The PoET solar telescope                              | June 2026 |
| <b>Physis Week 2025</b> , Python for data analysis                                                                                                   | Sept 2025 |
| <b>ENAA XXXV</b> <a href="#">↗</a> , A Quest to Detect a Star Velocity: Radial Velocity Extraction and Associated Challenges                         | Sept 2025 |
| <b>EPRV6</b> <a href="#">↗</a> , A systematic bias in template-based RV extraction algorithms                                                        | July 2025 |
| <b>Planets throughout the Habitable Zone</b> <a href="#">↗</a> , A systematic bias in template-based RV extraction algorithms                        | June 2025 |
| <b>PoET workshop 2</b> <a href="#">↗</a> , 2 talks: PoET observation software; Observing strategies and their role in the improvement of RV analysis | Nov 2024  |
| <b>TOI III</b> <a href="#">↗</a> , Approaches for RV extraction – s-BART and the first steps towards a fully Bayesian model                          | July 2023 |
| <b>EPRV 5</b> <a href="#">↗</a> , Towards a fully Bayesian RV extraction model                                                                       | Mar 2023  |
| <b>Exoplanets 4 EPRV splinter</b> , sBART application to the ESPRESSO WG1 targets                                                                    | Feb 2023  |
| <b>ESPRESSO science team meeting</b> , s BART a semi Bayesian implementation of template matching for precise Radial Velocities                      | May 2022  |

## Seminars

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| <b>Astronomical Institute of the Czech Academy of Sciences</b> , Avenues and challenges for radial velocity extraction – s-BART & PoET                                            | Aug 2025 |
| <b>Astronomy department, Uni. of Geneva</b> , Avenues for radial velocity extraction – s-BART & PoET                                                                              | Jan 2025 |
| <b>ESA research seminar, European Space Agency (Madrid)</b> , Towards an improvement in the characterisation of stellar radial velocities                                         | May 2024 |
| <b>Café com Física</b> <a href="#">↗</a> <b>Departamento de Física da Universidade de Coimbra</b> , The (radial) velocity of stars - detection and characterisation of exoplanets | Mar 2024 |

## Supervision

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| <b>Clustering of spectral lines based on their parameters</b> , Co-supervisor                                  | MsC thesis<br>Feb 2026 – present              |
| <b>Improved modelling of CCF profiles for activity-insensitive radial-velocity extraction</b> , Supervisor     | Undergraduate project<br>Feb 2026 – present   |
| <b>Measuring Zeeman Broadening in Stellar Spectra</b> , Co-supervisor                                          | Undergraduate project<br>July 2026 – present  |
| <b>Transforming H-alpha images of the Sun in astronomical seeing for the PoET solar telescope</b> , Supervisor | Undergraduate project<br>Feb 2025 – July 2025 |
| <b>Looking at the Sun, finding other Earths - Identification of Solar Regions</b> , Co-supervisor              | MsC thesis<br>Oct 22 - Nov 23                 |
| <b>Development of a GUI interface for the exploitation of data from the s-BART pipeline</b> , Co-supervisor    | Undergraduate project<br>Feb 2024 – July 2024 |
| <b>Solar-to-sky coordinate conversion for PoET operations</b> , Co-supervisor                                  | Undergraduate project<br>Feb 2024 – July 2024 |
| <b>Exoplanet detection through a new model for the correction of instrumental effects</b> , Co-supervisor      | Undergraduate project<br>Feb 2024 – July 2024 |

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| <b>Development of a tool for the normalization of stellar spectra - application to ESPRESSO data</b> , Co-supervisor of BII project with ref <a href="#">CIAAUP-03/2023-BII</a> | Funded project<br>Feb 2024 – July 2024    |
| <b>A new activity proxy for finding other Earths</b> , Co-supervisor                                                                                                            | Bachelors project<br>Nov 2022 – Feb 2023  |
| <b>An analysis of the performance of CHEOPS mission pipelines - the DRP and archi</b> , Co-supervisor                                                                           | Bachelors project<br>Feb 2020 – June 2020 |

## First Author Publications

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| <b>ASTRA: A Python Package for Cross-Instrument Stellar and Telluric Template Construction</b>                     | 2026 |
| André M. Silva et al; (JOSS, <a href="#">10.21105/joss.09413</a> )                                                 |      |
| <b>A systematic bias in template-based radial velocity extraction algorithms</b>                                   | 2025 |
| André M. Silva et al; (A&A, <a href="#">10.1051/0004-6361/202554955</a> )                                          |      |
| <b>A novel framework for semi-Bayesian radial velocities through template matching</b>                             | 2022 |
| André M. Silva et al; (A&A, <a href="#">10.1051/0004-6361/202142262</a> )                                          |      |
| <b>archi: pipeline for light curve extraction of CHEOPS background stars</b>                                       | 2020 |
| André M. Silva et al; (Monthly Notices of the Royal Astronomical Society, <a href="#">10.1093/mnras/staa1443</a> ) |      |

## Selected Co-authored Publications

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| <b>Gas-depleted planet formation occurred in the four-planet system around the red dwarf LHS 1903</b>                                            | 2026 |
| Wilson et al; (Science, <a href="#">10.1126/science.adl2348</a> )                                                                                |      |
| <b>A comprehensive study on radial velocity signals using ESPRESSO: Pushing precision to the 10 cm/s level</b>                                   | 2025 |
| Figueira et al; (A&A, <a href="#">10.1051/0004-6361/202553869</a> )                                                                              |      |
| <b>PoET: the Paranal solar ESPRESSO Telescope</b>                                                                                                | 2025 |
| Santos et al; (Published in The Messenger vol. 194, <a href="#">10.18727/0722-6691/5381</a> )                                                    |      |
| <b>TOI-512: Super-Earth transiting a K-type star discovered by TESS and ESPRESSO</b>                                                             | 2025 |
| Rodrigues et al; (A&A, <a href="#">10.1051/0004-6361/202452887</a> )                                                                             |      |
| <b>KOBE-1: The first planetary system from the KOBE survey: Two planets likely residing in the sub-Neptune mass regime around a late K-dwarf</b> | 2025 |
| Balsalobre-Ruza et al; (A&A, <a href="#">10.1051/0004-6361/202452631</a> )                                                                       |      |
| <b>TESS and ESPRESSO discover a super-Earth and a mini-Neptune orbiting the K-dwarf TOI-238</b>                                                  | 2024 |
| Suárez Mascareño et al; (A&A, <a href="#">10.1051/0004-6361/202348958</a> )                                                                      |      |
| <b>A sub-Earth-mass planet orbiting Barnard's star</b>                                                                                           | 2024 |
| González Hernández et al; (A&A, <a href="#">10.1051/0004-6361/202451311</a> )                                                                    |      |
| <b>Two temperate Earth-mass planets orbiting the nearby star GJ1002</b>                                                                          | 2023 |
| Mascareño et al; (A&A, <a href="#">10.1051/0004-6361/202244991</a> )                                                                             |      |
| <b>An unusually low-density super-Earth transiting the bright early-type M-dwarf GJ 1018 (TOI-244)</b>                                           | 2023 |
| Castro-González et al; (A&A, <a href="#">10.1051/0004-6361/202346550</a> )                                                                       |      |
| <b>A candidate short-period sub-Earth orbiting Proxima Centauri</b>                                                                              | 2022 |
| Faria et al; (A&A, <a href="#">10.1051/0004-6361/202142337</a> )                                                                                 |      |

**The KOBE experiment: K-dwarfs Orbiting By habitable Exoplanets: Project goals, target selection, and stellar characterization**

2022

Lillo-Box et al; (A&A, [10.1051/0004-6361/202243898](https://doi.org/10.1051/0004-6361/202243898) [↗](#))

## Organization

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**Coordinator of internal working group focused on modelling time-series for exoplanet detection and characterisation** 2025 - now

**Organizer of the scientific seminars of CAUP** September 2025 - now

**Co-organizer of PoET's Working Group 1 - Radial velocities** 2025 - now

**Coordinator and lecturer of advanced course "Python for astronomers"** 2021 - now

**SoC of Dias da Física 2025** 2025

**Part of the Local Organization committee of EPRV6** 2025

## Outreach

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**Outreach talk in Jornadas Espaciais 2026 (Lisbon, Portugal); "Exoplanetas e Mundos distantes"** May 2026

**Outreach talk in COMICCON 2026 (Portugal); "PoET - um telescópio solar português"** Apr 2026

**Ignite sessions; 'Torres Vedras [↗](#) Ílhavo [↗](#) Armamar'** 2023-2024

**Outreach talks to highschool students: Espaço vai à Escola - Descoberta de outra Terra - deteção de planetas fora do sistema solar** 2022-2025

2022 (3 talks); 2023 (5 online; 5 in-person); 2024 (6 online; 4 in-person); 2025 (6 in-person; 17 online)